Program: **Diploma in Electronics and Computer Engineering**	
Course Code: **6501A**	Course Title: **Entertainment Electronics**
Semester: **6**	Credits: **4**
Course Category: **Open elective**	
Periods per week: **4 (L:4 T:0 P:0)**	Periods per semester: **60**

Course Objectives:

- The course aims to provide broad knowledge on various industry standards, algorithms and technologies used to carry out digital audio and video broadcasting in the infotainment industry.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge in Electrical and Electronics Engineering concepts	2041	Basic Electronics	*2*
	2031	Fundamentals of Electrical and Electronics Engg.	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Understand packetized streaming of digital media happens in the field of infotainment industry.	12	Understanding
CO2	Explain the critical aspects of DVB standards used for media broadcasting in infotainment industry.	16	Understanding
CO3	Explain the critical aspects of DAB standards used for media broadcasting in infotainment industry.	13	Understanding
CO4	Understand modern display technologies for video reproduction.	17	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3	3					
CO2	3	3		2			
CO3	3	3		2			
CO4	3	3					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of the Experiment	Duration (Hours)	Cognitive Level
CO1	**Understand packetized streaming of digital media happens in the field of infotainment industry.**		
M1.01	Brief Review of Analog Television:	4	Understanding
M1.02	Explain Digital media streaming:	8	Understanding
Contents: ● **Brief Review of Analog Television**: Scanning, Horizontal and Vertical Synchronization, Color information, Transmission methods. NTSC and PAL standards. ● **Digital media streaming**: Packetized elementary stream of audio-video data, MPEG data stream, MPEG-2 transport stream packet, Accessing a program, scrambled programs, program synchronization. PSI, Additional (Network information and service description)information in data streams for set-top boxes.			
CO2	**Explain the critical aspects of DVB standards used for media broadcasting in infotainment industry.**		
M2.01	Explain satellite TV broadcasting	4	Understanding
M2.02	Explain cable TV broadcasting	4	Understanding
M2.03	Explain terrestrial TV broadcasting	4	Understanding
M2.04	Illustrate broadcasting for Handheld devices	4	Understanding
	Series Test – I	1	
Contents: ● **Satellite TV broadcasting** –DVB-S Parameters, DVB-S Modulator, DVB-S set-top box, DVB-S2.			

• **Cable TV broadcasting** – DVB-C Standard, DVB-C Modulator, DVB-C set-top box. • **Terrestrial TV broadcasting** – DVB-T Standard, DVB-T Modulator, DVB-T Carriers and System Parameters, DVB-T receiver. • **Broadcasting for Handheld devices** – DVB-H Standard DVB tele-text, DVB subtitling system.			
CO3	**Explain the critical aspects of DAB standards used for media broadcasting in infotainment industry**		
M3.01	Explain Digital Audio Broadcasting (DAB)	6	Understanding
M3.02	Describe Digital Radio Mondiale (DRM)	7	Understanding
Contents: • **Digital Audio Broadcasting (DAB)**: Comparison of DAB with DVB. Physical layer of DAB. DAB Modulator, DAB Data Structure, DAB single frequency networks, Data broadcasting using DAB. • **Digital Radio Mondiale (DRM)**: Transmitter and receiver, Data rates.			

CO4	**Understand modern display technologies for video reproduction.**		
M4.01	Explain display Technology	10	Understanding
M4.02	Describe television of future	7	Understanding
	Series Test – II	1	
Contents: • **Display Technology**: Block diagram of video reproduction system in a TV, Cathode Ray tubes, Basic principle of Plasma displays, LCD displays, Light-emitting diode displays, Field emission displays, Organic light emitting device displays. • **Television of future:** Holographic TV, Virtual Reality, Augmented Reality.			

Text / Reference

T/R	Book Title/Author
T1	W. Fischer, **Digital Video and Audio Broadcasting Technology**: A Practical Engineering Guide (Signals and Communication Technology), Springer, 2020
T2	Lars-Ingemar Lundström, **Understanding Digital Television An Introduction to DVB Systems with Satellite, Cable, Broadband and Terrestrial TV**, Focal Press, Elsevier,2006.
T3	K F Ibrahim, **Newnes Guide to Television and Video Technology**, Newnes, 2007.

T/R	Book Title/Author
T4	Jiun-Haw Lee, David N. Liu, Shin-Tson Wu, **Introduction to Flat Panel Displays**, Wiley, 2008.
R1	Wolfgang Hoeg, Thomas Lauterbach, **Digital audio broadcasting: principles and applications of DAB, DAB+ and DMB**, Wiley, 2009.
R2	John Watkinson, **Introduction to Digital Audio**, Focal Press, 1994.
R3	John Watkinson, Introduction to Digital Video, Focal Press, 2001.

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/106/105/106105166/
2	https://www.raspberrypi.org/blog/getting-started-with-iot/
3	https://learn.adafruit.com/category/internet-of-things-iot
4	https://www.arduino.cc/en/IoT/HomePage
5	http://esp32.net/